

A STUDY AND ANALYSIS ON CRYPTOCURRENCY TRADING

Submitted by

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Under the Guidance of

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ABSTRACT :

This study analyzes algorithms used for predicting cryptocurrency trading outcomes using available datasets. It explores machine learning and statistical models to be implemented very accurately and to perform diversified calculations for the predictions. The research highlights challenges like data scarcity, market volatility, and algorithmic biases. Through comparative analysis, it identifies efficient approaches to enhance predictive accuracy, aiding traders and investors in better decision-making.

The algorithms implemented:

- **RANDOM FOREST ALGORITHM**
- **XGBOOST ALGORITHM**
- **LINEAR REGRESSION ALGORITHM**
- **POLYNOMIAL REGRESSION ALGORITHM**

By these algorithm the overall structure and the accuracy variance can be differentiated by the amount of components which we include on these algorithms and the overall graphical visualization can also be differentiated.

OBJECTIVE :

- To analyze various algorithms used for cryptocurrency price prediction.
- To evaluate the effectiveness of machine learning and statistical models.
- To identify challenges such as data scarcity and market volatility.
- To compare different prediction techniques for accuracy and reliability.
- To develop insights that enhance decision-making in cryptocurrency trading.



ALGORITHMS AND THEIR OUTCOMES:

★ RANDOM FOREST ALGORITHM:

- Used for cryptocurrency price prediction due to accuracy and resilience.
- Builds multiple decision trees and aggregates results for better accuracy.
- Trained on random subsets to reduce overfitting and improve generalization.
- Handles large datasets with noisy and missing values.
- Uses **SMA_7**, **EMA_7**, and other parameters for scalable results.
- **Accuracy** improved from **8523** to **8982** with optimized parameters.
- **SMA_21**, **EMA_21**, **RSI**, **MACD**, and **Bollinger Bands** tested for better predictions.

Feature	Previous Model	Current Model
Indicators Used	SMA_7, SMA_21, EMA_7, EMA_21, RSI, MACD, Bollinger Bands	Only SMA_7 & EMA_7
Hyperparameter Tuning	✔ Used <code>GridSearchCV</code>	✗ Not used (fixed params)
Features Used	13 Features	6 Features
Random Forest Parameters	Tuned via GridSearchCV	<code>n_estimators=100,</code> <code>max_depth=10</code>
Model Complexity	Higher (more features & tuning)	Lower (simpler model)

XGBOOST ALGORITHM :

- Used for cryptocurrency price prediction due to high speed and accuracy.
- Builds multiple decision trees sequentially to correct errors and enhance performance.
- Optimized for handling large datasets efficiently with regularization to reduce overfitting.
- Processes noisy and missing data, making it suitable for volatile markets.
- Uses **SMA_7** and **EMA_7** as key parameters for prediction.
- Splits the dataset into **80%** for training and **20%** for testing for effective evaluation.
- **Accuracy** improved from **8960** to **9051**, making it highly scalable for predictions.

Hyperparameter	Original Value	Modified Value	Impact
learning_rate	[0.01, 0.1, 0.2]	[0.01, 0.05, 0.1]	Lowering the upper range (from 0.2 → 0.1) makes training more stable. A 0.05 step is added for fine-tuning.
max_depth	[3, 5, 7]	[3, 6, 9]	Increased upper depth (7 → 9) allows deeper trees to capture complex patterns, improving accuracy but increasing overfitting risk.
n_estimators	[100, 500, 1000]	[500, 1000, 1500]	Removed low values (100), increasing 1500 , making the model more powerful but computationally expensive.
subsample	[0.6, 0.8]	[0.6, 0.8, 1.0]	Added 1.0 to allow full dataset training instead of random sampling, reducing bias.
colsample_bytree	[0.6, 0.8]	[0.6, 0.8, 1.0]	Added 1.0 to let the model consider all features, improving learning.
gamma	[0, 0.1]	[0, 0.1, 0.2]	Added 0.2 to further control tree pruning, preventing overfitting.

LINEAR REGRESSION ALGORITHM:

THE OBJECTIVE :

- Used for cryptocurrency price prediction due to its simplicity and efficiency.
- Establishes a linear relationship between input features and price trends.
- Utilizes **SMA, EMA, RSI, RFE, R², RMSE, MAE, Accuracy**, and **High L2 regularization** for improved performance.
- Reduces overfitting and enhances model generalization with regularization techniques.
- Splits the dataset into **80%** for training and **20%** for testing for effective evaluation.
- Processes noisy data but struggles with extreme market volatility.
- **Accuracy** improved progressively from **3679** → **5541** → **9681**, making it highly scalable for cryptocurrency predictions.

Comparison Table

Feature	Code 1: Dummy Regressor	Code 2: Ridge Regression (High Alpha)	Code 3: Optimized Linear Regression
Features Used	Open, High, Low, Volume	Open, High, Low, Volume	Open, High, Low, Volume, SMA, EMA, RSI
Feature Engineering	✗ None	✗ None	✓ SMA, EMA, RSI
Feature Scaling	✗ No	✓ StandardScaler	✓ MinMaxScaler
Feature Selection	✗ No	✗ No	✓ Recursive Feature Elimination (RFE)
Train-Test Split	80% Train, 20% Test	50% Train, 50% Test	80% Train, 20% Test (Sequential)
Model Used	Dummy Regressor (Mean)	Ridge Regression ($\alpha=1000$)	Linear Regression
Regularization	✗ No	✓ High L2 Regularization	✗ No
Predictions Scaling Back	✗ No	✗ No	✓ Yes (Inverse Transform)
Evaluation Metrics	RMSE, R ² , Accuracy	RMSE, R ² , Accuracy	RMSE, R ² , MAE, Accuracy
Graphical Output	✗ No	✗ No	✓ Yes (Actual vs Predicted Plot)
Strengths	Baseline comparison	Tests extreme regularization	Optimized with feature engineering and selection
Weaknesses	Not predictive	Over-regularized (underfits)	More complex

POLYNOMIAL REGRESSION ALGORITHM :

THE OBJECTIVE:

- Used for cryptocurrency price prediction by capturing nonlinear relationships.
- Extends Linear Regression by adding polynomial features for better trend fitting.
- Utilizes **MinMax Scaler, Standard Scaler, RidgeCV (L2 Regularization), and RFE** for optimized performance.
- Reduces overfitting and improves model generalization with **RidgeCV regularization**.
- Handles complex patterns in data but may struggle with extreme market fluctuations.
- Accuracy improved progressively from **-3.6301** → **9166** → **9597**, making it highly scalable for cryptocurrency predictions.

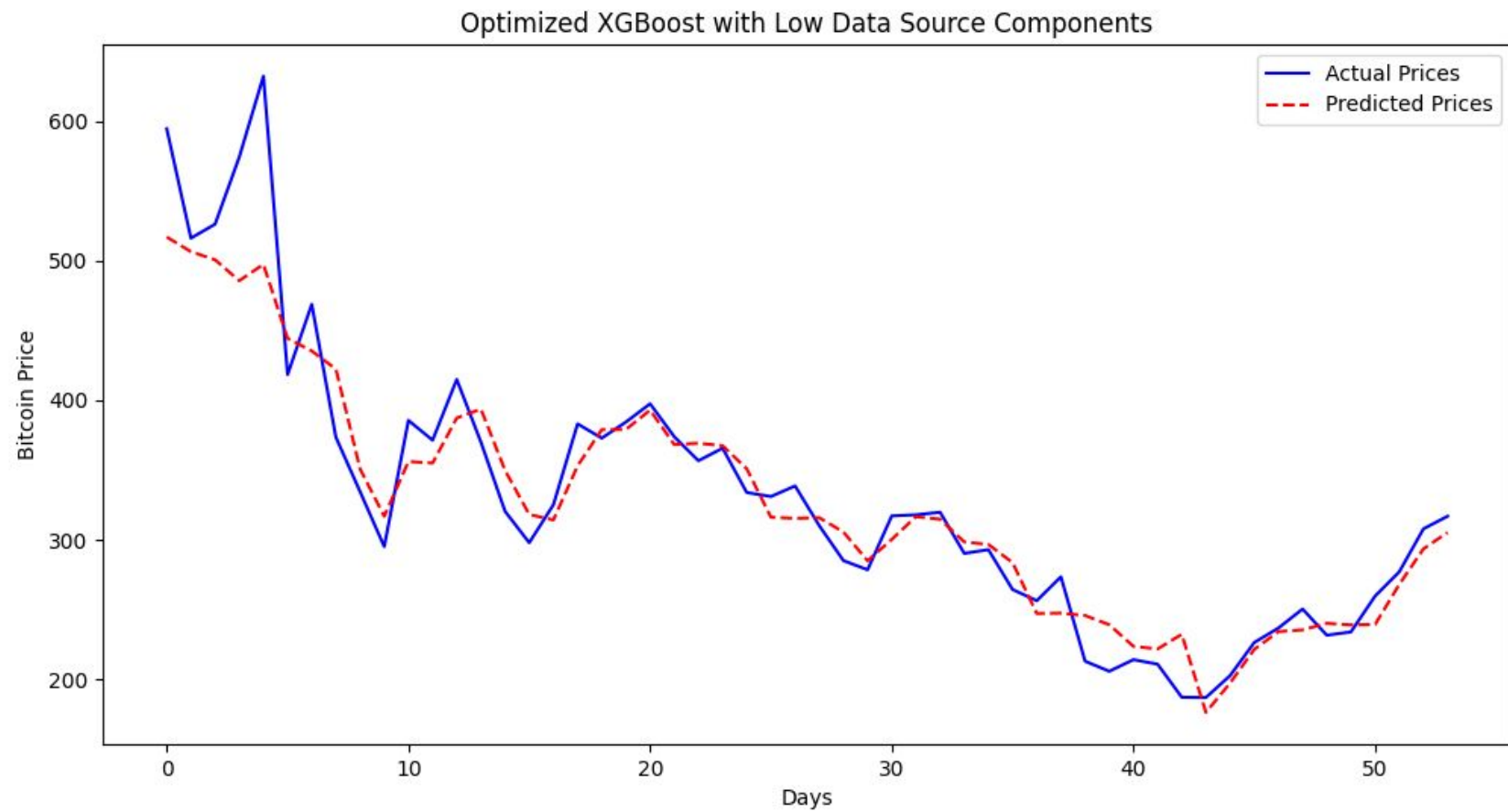
Key Differences in Tabular Format

	Code 1: RFE + Poly (Degree = 3)	Code 2: RFE + Poly (Degree = 2)	Code 3: RidgeCV + Poly (Degree = 4)
Feature			
Feature Selection	RFE (12 features)	RFE (8 features)	No selection (all features used)
Polynomial Degree	3	2	4
Regularization	No regularization	No regularization	RidgeCV (L2 Regularization)
Feature Scaling	MinMaxScaler	MinMaxScaler	StandardScaler
Complexity	Moderate	Low	High
Overfitting Risk	High	Moderate	Low (due to RidgeCV)
Best For	Medium-sized datasets	Small datasets	Large datasets with feature interactions
Performance Control	Feature Selection via RFE	Feature Selection via RFE	Regularization via Ridge

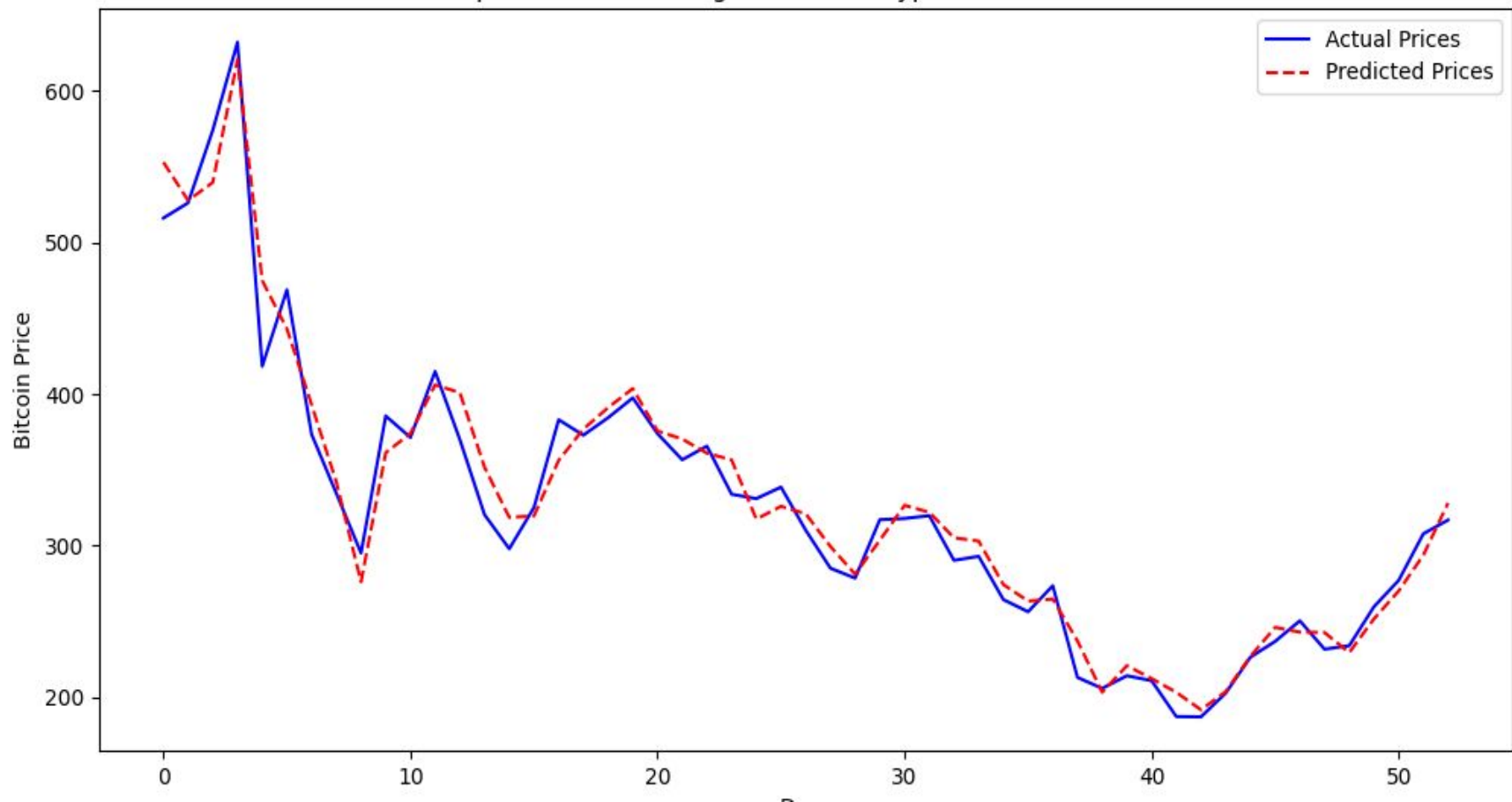
REQUIRED COMPONENTS :

Category	Components	Purpose
Algorithms	Random Forest, XGBoost, Linear Regression, Polynomial Regression	Used for predicting cryptocurrency prices
Feature Selection	RFE (Recursive Feature Elimination)	Selects important features for better accuracy
Scaling Techniques	MinMax Scaler, Standard Scaler	Normalizes data for improved model performance
Regularization	RidgeCV (L2 Regularization), HighL2	Prevents overfitting and enhances generalization
Evaluation Metrics	R ² (R-Squared), RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), Accuracy	Measures model performance and error rate
Technical Indicators	SMA (Simple Moving Average), EMA (Exponential Moving Average), RSI (Relative Strength Index), MACD, Bollinger Bands	Helps in trend analysis and feature extraction
Data Processing	Handling Missing Values, Noise Reduction	Ensures clean and reliable dataset for predictions
Hardware Requirements	CPU: Intel i5/i7 or AMD equivalent, RAM: 8GB+ (16GB recommended), Storage: SSD (256GB+), GPU: NVIDIA/AMD (for faster computations)	Ensures efficient computation and model training
Python Modules	NumPy, Pandas, Scikit-learn, XGBoost, Matplotlib, Seaborn, Statsmodels, TensorFlow/Keras (if deep learning is used)	Supports data processing, model training, and visualization

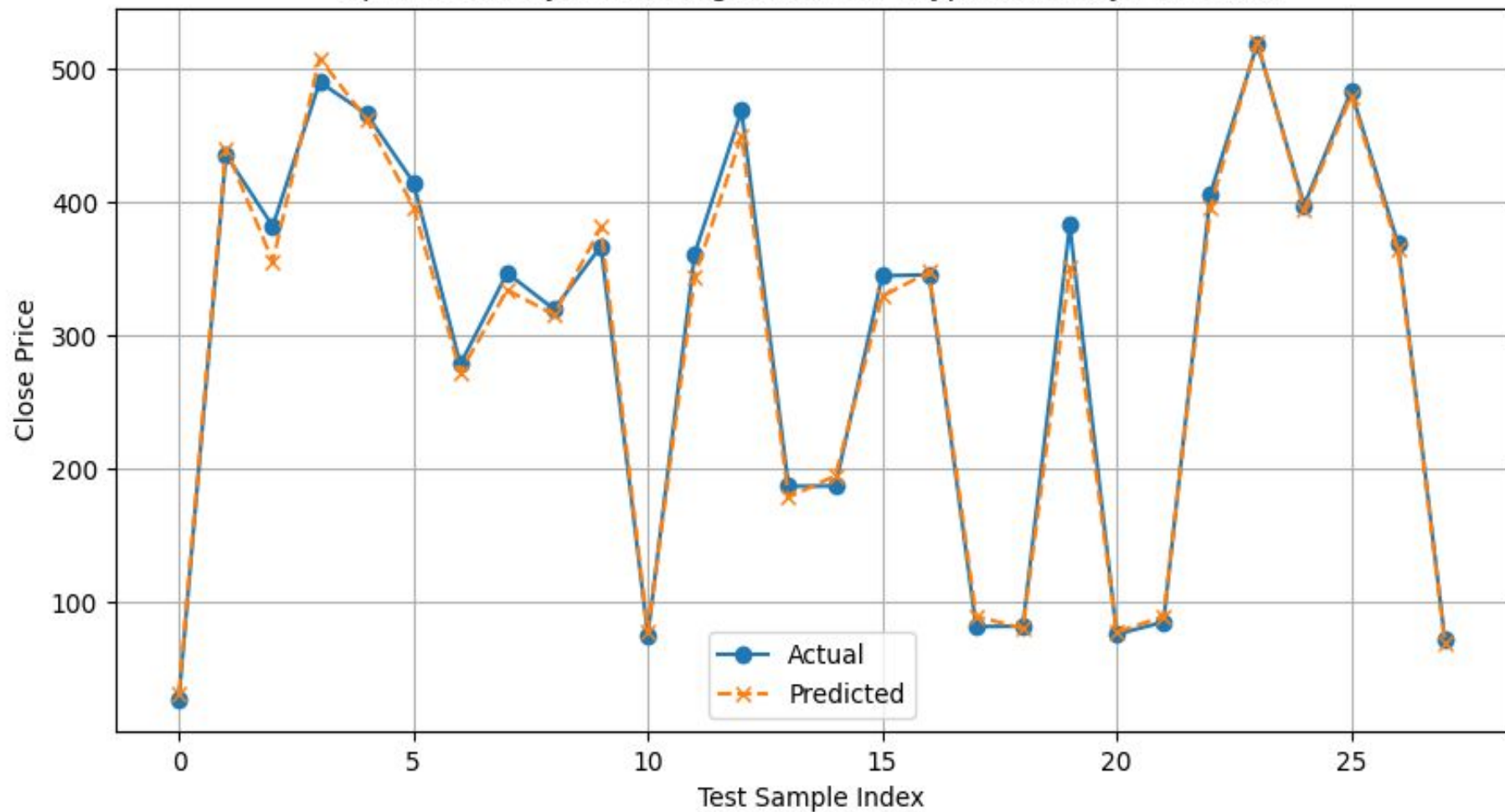
VISUALISATION :



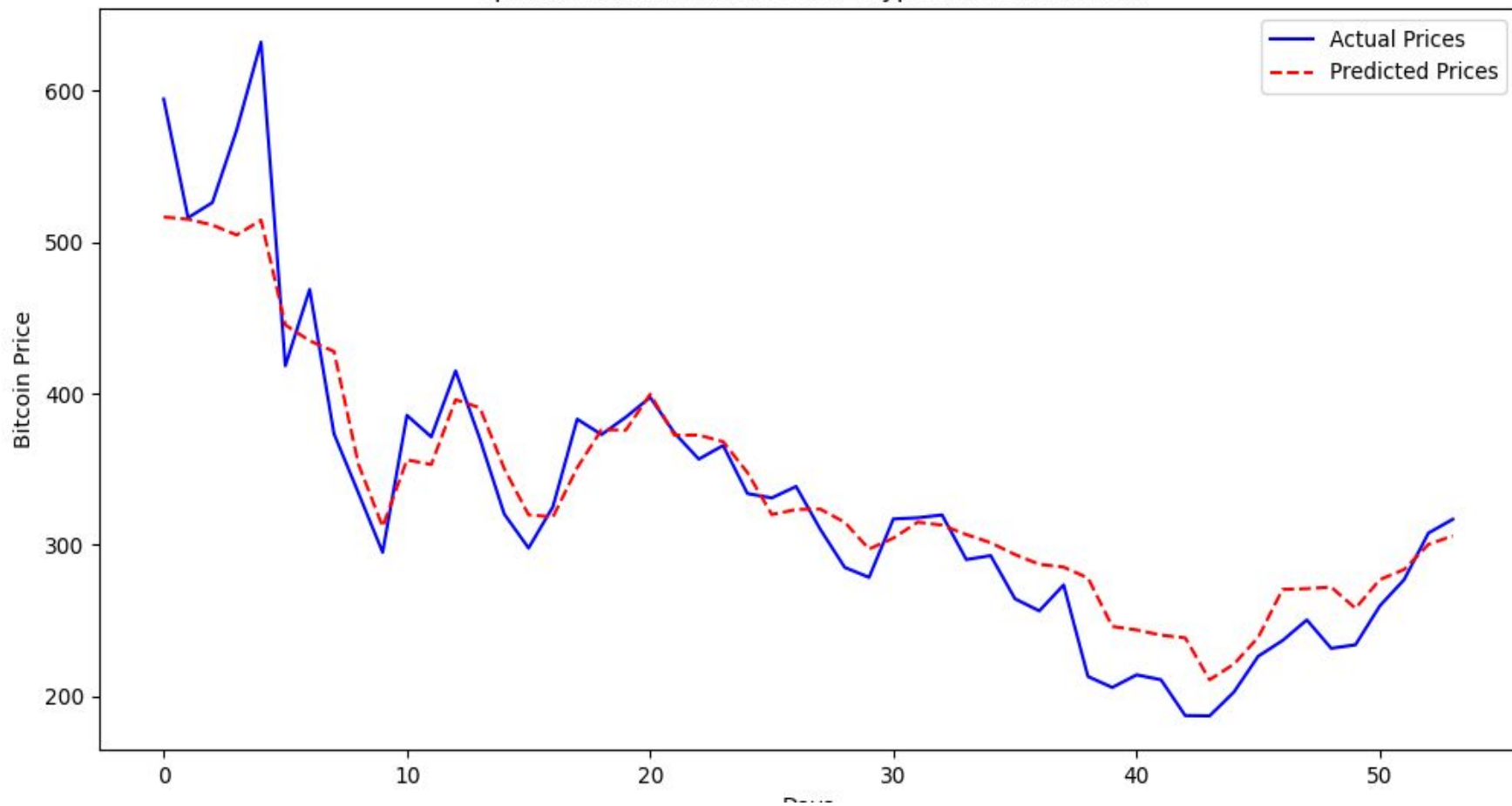
Optimized Linear Regression for Crypto Price Prediction



Optimized Polynomial Regression for Cryptocurrency Prediction



Optimized Random Forest for Crypto Price Prediction



= RESTART: C:\Users\nithi\AppData\Local\Programs\Python\Python310\xboost_algorithm_sma_and_ema_9051.py

Optimized XGBoost Model Performance:

Accuracy (R^2 Score): 0.9051

MAE: 20.7885

RMSE: 30.9065

= RESTART: C:\Users\nithi\AppData\Local\Programs\Python\Python310\randomforest_low_best_8982.py

Optimized Random Forest Model Performance:

Accuracy (R^2 Score): 0.8982

MAE: 24.0259

RMSE: 32.0101

= RESTART: C:\Users\nithi\AppData\Local\Programs\Python\Python310\linear_regression_requiredcomponents_9681.py

Optimized Linear Regression Model Performance:

Accuracy (R^2 Score): 0.9681

MAE: 12.8731

RMSE: 16.8716

= RESTART: C:\Users\nithi\AppData\Local\Programs\Python\Python310\polynomial_ridge_regression_0.9597.py

OPTIMIZED POLYNOMIAL REGRESSION MODEL PERFORMANCE:

Accuracy: 0.9597

RMSE: 12.1775

R2 Score: 0.9937



Thank
you

